Ivana Lovric

Software Developer | Engineer in Mobility Services & AI

Analytical full-stack developer with an engineering mindset

Driven software developer combining a strong academic background in engineering (Master's degree in Wood Technology) with a modern and powerful tech stack. Unique ability to navigate between complex algorithms and user-focused frontend - with deep technical understanding and business relevance. Experience in advanced machine learning, cloud-based architecture, and large-scale integrations. A solution-oriented team player who thrives in agile environments where technology meets the mobility challenges of tomorrow.

LinkedIn

GitHub

Portfolio

SKILLS & TOOLS

- **Languages & Frameworks:** Java (21, Spring Boot), Python (Machine Learning/AI), TypeScript, React (19), JavaScript, C++, SQL, HTML5, CSS
- **Technical Environment:** AWS (Lambda, DynamoDB, AppSync), GraphQL (Apollo Client), REST API, MySQL, JDBC, PostgreSQL, Jupyter Lab, OpenAI API
- **Development Tools:** Git, GitHub, GitHub Actions (CI/CD), Vite, Recharts, Linear, Jira, Trello, IntelliJ IDEA, PyCharm
- **Methodologies & Expertise:** Agile Development (Scrum), End-to-end ML Pipelines, Feature Engineering, Data Extraction (LLM/Regex), System Architecture, UI/UX Focus, Test-Driven Development (JUnit, Mockito)

WORK EXPERIENCE

AI Developer / RAG & Search Systems (Internship)

April 2026 - Present

Company/Project: ZoCom AB / YH Course Matcher

Assignment: Development of an AI-based tool that matches teacher CVs, competencies and free text against relevant vocational education (YH) courses, responsible for the full chain from data collection and text extraction to semantic search, ranking and user interface.

- Building pipeline flows for PDF text extraction with PyMuPDF and OCR fallback via OpenAI, with normalisation to CSV for a searchable data source.
- Implementing semantic search and hybrid ranking with embeddings and vector indexing in Pinecone, combined with lexical overlap for high accuracy.
- Developing a Streamlit-based application with filtering, grouping and export functions, being prepared for deployment on Render.

Tools/Environments: Python, Streamlit, OpenAI API, Pinecone, PyMuPDF, Git, Render.

AI Developer / ML Engineer (Internship)

Feb 2026 - Mar 2026

Company/Project: ZoCom AB / Swedish Agency for Higher Vocational Education (MYH)

Assignment: Development of an intelligent AI agent for automated analysis and strategic ranking of vocational education applications.

- Architecting and implementing an end-to-end ML pipeline for large-scale data processing.
- Developing advanced data extraction logic from complex PDF documents using Large Language Models (GPT-4o) and Regex.
- Responsible for the full process from raw data to finished ranking model, including feature engineering, model training and cross-validation.
- Developing visualisations and continuously improving scoring logic in matplotlib/seaborn to strengthen decision support quality and reduce uncertainty in assessments.

Tools/Methods: Python, Jupyter Lab, OpenAI API, scikit-learn, matplotlib, seaborn, pypdf, pdfplumber.

Full-Stack Developer (Internship)

Jan 2026 - Feb 2026

Company/Project: ZoCom AB / Azomo

Assignment: Further development and optimisation of Azomo - a modern and scalable educational platform.

- Designed and implemented type-safe React components in TypeScript with a focus on complex dependencies and sorting algorithms.
- Ensured system performance through implementation of a doubly linked list for efficient node manipulation and automatic course navigation.
- Integrated the frontend against a serverless backend via GraphQL (Apollo Client) and AWS infrastructure (Lambda, DynamoDB, AppSync).
- Contributed to the maintenance and development of Azomo Business - a cloud-based SaaS platform - with continuous testing and quality assurance in a monorepo environment to improve stability and reduce regression risk.

Tools/Environments: React, TypeScript, GraphQL, AWS (Lambda, DynamoDB, AppSync), Vite, Git.

City Center One - Information Desk Associate (student job) 2018 - 2019

Assisted visitors at the information desk and ensured high service quality in a large shopping centre environment.

NCVVO - Examination Administrator (student job) 2017 - 2018

Planning and conducting national graduation examinations for high school students on behalf of Croatia's National Centre for External Evaluation of Education.

Student Jobs - Müller, Milsped (H&M), Kuehne+Nagel, FOREO 2016 - 2017

Alongside my studies I worked in retail, warehouse logistics and production - a good balance between learning and developing professional work habits within international companies.

T.O. Ruza - Warehouse & Trade Assistant (family business) 2009 - 2013

Contributed to the family-owned trading company in import, export and retail - from the start of my secondary education, which gave me an early practical understanding of business operations and working life.

SELECTED PROJECTS

Trainy - Graduation Project (Full-Stack) March 2026

Assignment: Development of a data-driven platform for real-time analysis of train punctuality, tailored for the mobility sector.

- Built a high-performance backend in Java 21 / Spring Boot with seamless integration against Trafikverket's open API.
- Developed an intuitive dashboard in React 19 that visualises delay statistics per station and train number using Recharts.

Chat Backend & System Support 2025

Assignment: Design of scalable system solutions for communication and data storage.

- Implemented a robust REST API in Java with a focus on quality assurance through unit tests and integration tests (JUnit, Mockito).
- Created a database-driven application with role-based access control via JDBC and MySQL.

EDUCATION

Software Developer for Mobility Services (400 YH credits) · Folkuniversitetet 2024 - Present

Master's Degree in Wood Technology (120 ECTS credits) · University of Zagreb 2018 - 2020

Bachelor's Degree in Wood Technology (180 ECTS credits) · University of Zagreb 2013 - 2018

Swedish as a Second Language, upper secondary level · MoA Lärcentrum 2021 - 2023

Languages: Swedish · English · Croatian

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References from ZoCom AB available upon request.